Yuta Oshima

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Education

Apr. 2025 – Present	Ph.D. in Computer Science at **The University of Tokyo** Advisor: Prof. Yutaka Matsuo
Apr. 2023 – Mar. 2025	M.S. in Computer Science at **The University of Tokyo**
Apr. 2019 – Mar. 2023	B.S. in Engineering at **The University of Tokyo**

Publications

[1] **Y. Oshima**, K. Onoda, Y. Iwasawa, M. Suzuki, Y. Matsuo, H. Furuta. *AutoRef: Harness Optimization for Agentic Multi-Reference Image Generation*. Preprint. [arXiv] [Code]

[2] **Y. Oshima**, M. Yoshimura, M. Suzuki, Y. Matsuo, H. Furuta. *VIF-Bench: Evaluating Visual Instruction Following in Multi-Reference Image Generation*. Preprint. [arXiv] [Code] [HuggingFace]

[3] **Y. Oshima**, D. Miyake, K. Matsutani, Y. Iwasawa, M. Suzuki, Y. Matsuo, H. Furuta. *MultiBanana: A Challenging Benchmark for Multi-Reference Text-to-Image Generation*. **CVPR 2026 Main**. [arXiv] [Code] [HuggingFace]

[4] **Y. Oshima**, Y. Iwasawa, M. Suzuki, Y. Matsuo, H. Furuta. *WorldPack: Dynamic Frame Compression for Long-context Video World Modeling*. **TMLR**. [arXiv] [OpenReview]

[5] **Y. Oshima**, M. Suzuki, Y. Matsuo, H. Furuta. *Inference-Time Text-to-Video Alignment with Diffusion Latent Beam Search*. **NeurIPS 2025**. [arXiv] [OpenReview] [Code] [Project]

[6] **Y. Oshima**, S. Taniguchi, M. Suzuki, Y. Matsuo. *SSM Meets Video Diffusion Models: Efficient Video Generation with Structured State Spaces*. **New Generation Computing**. [DOI] [arXiv]

[7] **Y. Oshima**, M. Suzuki, Y. Matsuo. *Enhancing Unimodal Latent Representations in Multimodal VAEs through Iterative Amortized Inference*. **New Generation Computing**. [DOI] [arXiv]

[8] K.Onoda, P.Parmas, H.Furuta, S.Nishimori, **Y.Oshima**, S.Taniguchi, Y.Matsuo. *Multi-Axis MaxK Reinforcement Learning for Representative Diversity in Text-to-Image Generation*. **WACV 2027**. [arXiv]

[9] H. Zhang, **Y. Oshima**, X. Du, C. Wang, I. Li, Y. Iwasawa, Y. Matsuo. *SMC-ITA: Sequential Monte Carlo Inference-Time Alignment for Video-to-Audio Generation*. **APSIPA ASC 2026**. [arXiv]

[10] R. Takahashi, R. Li, **Y. Oshima**, S. Kuno, R. Morita, I. Sukeda. *AKITalk: Audio-Implicit Keypoints for Identity-Preserving Talking-Head Video Synthesis*. **MMAsia 2025 Grand Challenge**, **3rd Place**. [DOI]

[11] S. Taniguchi, K. Harada, G. Minegishi, **Y. Oshima**, S. C. Jeong, G. Nagahara, T. Iiyama, M. Suzuki, Y. Iwasawa, Y. Matsuo. *ADOPT: Modified Adam Can Converge with Any β_2 with the Optimal Rate*. **NeurIPS 2024**. [arXiv] [OpenReview] [Code]

[12] T. Kamijo, T. Iiyama, **Y. Oshima**, G. Venture, T. Matsushima, Y. Matsuo, Y. Iwasawa. *Tactile In-Hand Pose Estimation through Perceptual Inference*. **IROS 2023 Workshop**, **Spotlight**. [PDF]

Work & Research Experience

Google PhD Fellow, Google Oct. 2026 – Sep. 2027

- Awarded the 2026 Google PhD Fellowship in Machine Perception.
- Conducted independent PhD research with the mentorship of Hiroki Furuta at Google DeepMind.

Research Intern, Sakana AI Oct. 2026 – Present

- Research on Video World Models.

Research Intern, AIRoA Oct. 2026 – Present

- Research on Robot Foundation Models.

Researcher, EQUES Inc. Jul. 2025 – May 2026

- Image and video generation for anime production within the GENIAC project, supported by Japan's Ministry of Economy, Trade and Industry (METI).

PhD Student, The University of Tokyo Apr. 2023 – Present

- Matsuo–Iwasawa Lab, advised by Prof. Yutaka Matsuo.

Awards & Honors

2026 **Google PhD Fellowship in Machine Perception**

2026 **Invited Review Article**, Journal of the Imaging Society of Japan

2026 **Overseas Travel Grant**, The Telecommunications Advancement Foundation

2025 **3rd Place**, The ACM MMAsia 2025 Grand Challenge: Multimodal Multiethnic Talking-Head Video Generation [DOI]

2025 **Research Funding**, The University of Tokyo Leadership Development Program for Ph.D.

2025 **Research Funding**, The Nippon Foundation HUMAI Program

2024 **Outstanding Presentation Award**, JSAI Annual Conference

2023 **Spotlight Presentation**, IROS 2023 Workshop on World Models and Predictive Coding in Cognitive Robotics [PDF]

Academic Service

Conference Reviewer: CVPR (2026), ECCV (2026), ICLR (2026, 2027), ICML (2026; silver reviewer), NeurIPS (2026), WACV (2027)

Journal Reviewer: TMLR, Artificial Intelligence Review